

Tharun Kumar Marada

☎ +46 793523196 | ✉ tharunlol15@gmail.com | 🌐 Portfolio | 🐙 GitHub

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production-ready machine learning systems, autonomous agent pipelines, and generative AI applications. I have designed and shipped end-to-end solutions across anomaly detection, churn prediction, RAG systems, and multi-agent orchestration using Python, SQL, LangChain, LangGraph, PyTorch, and cloud infrastructure, always with a focus on reliability, explainability, and real business impact.

TECHNICAL SKILLS

AI & LLMs | LangChain, LangGraph, RAG, Agentic AI, ChromaDB, OpenAI API, Anthropic API, NLP, Prompt Engineering

Machine Learning | Scikit-learn, XGBoost, PyTorch, TensorFlow, Anomaly Detection, SHAP, Feature Engineering, Model Evaluation

Data and Engineering | Python (advanced), SQL, PostgreSQL, FastAPI, Docker, Git, Pandas, NumPy, Streamlit, CI/CD basics

Cloud | AWS (EC2, S3, SageMaker), Azure ML, GitHub Actions, Vercel

EDUCATION

MSc Computer Science Blekinge Institute of Technology, Sweden

Aug 2024 – Jun 2026

BTech Computer Science and Engineering JNTU, India

2020 – 2024

WORK EXPERIENCE

Machine Learning Intern

Summer 2025

BIST Technologies

- Built end-to-end anomaly detection and classification pipelines in Python and Scikit-learn on real operational data, with automated quality validation steps built into each pipeline stage
- Designed reproducible, well-documented workflows so systems could be maintained and extended by the wider team without reverse-engineering prior decisions
- Translated model outputs and performance trade-offs into plain-language summaries for non-technical stakeholders, turning results into decisions rather than just reports

Student Ambassador

Jan 2025 – Jul 2025

Blekinge Institute of Technology, Sweden

- Guided new international students through campus onboarding, led campus tours, and provided hands-on support with campus technology and systems
- Organised and led Indian cultural events on campus, building community among international students and contributing to a more inclusive campus environment

KEY PROJECTS

Multi-Agent AI Research Generator

[LINK](#)

- Built a 4-agent autonomous pipeline using LangGraph planner, searcher, extractor, writer — producing cited PDF reports from 15 to 20 live sources in under 90 seconds
- Engineered inter-agent state handoff logic and per-stage error handling so the system runs reliably end to end without human intervention

Customer Churn Prediction and BI Dashboard

[LINK](#)

- Trained XGBoost on 7,043 customer records achieving AUC-ROC 0.838, identified \$1.67M annual revenue at risk, and applied SHAP to surface the top churn drivers per prediction
- Built a SQL data model with cohort analysis views and deployed a live Streamlit dashboard with real-time risk scoring for individual customer profiles

Document Intelligence RAG System

[LINK](#)

- Built a retrieval-augmented generation pipeline using LangChain and ChromaDB that ingests any PDF and returns cited natural language answers in under 3 seconds
- Designed for zero reconfiguration across document types — financial reports, legal contracts, research papers — with grounded cited responses at every query

LLM-Based Intrusion Detection System

[LINK](#)

- Trained a Random Forest classifier on 25,000 real network connections achieving 99.73% F1 and 0.43% false positive rate through systematic class imbalance handling
- Layered LLaMA 3 to generate plain-English explanations per alert, making the system interpretable to non-specialist analysts rather than just returning a score